COHOMOLOGICAL DESCENT FOR FPPF HYPERCOVERINGS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-compare-simplicial-objects-fppf-etale, lemma-direct-image-morphism-simplicial-sites, lemma-hypercovering-X-simple-descent-bounded-abelian; external tool from More on Cohomology of Spaces.

Editorial note. The label lemma-cohomological-descent-for-fppf-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Comparison diagram. Same unit, Lemma lemma-compare-simplicial-objects-fppf-etale. Commutative diagram relating a and a_fppf used for the second equality.
- fppf hypercovering descent on components. More on Cohomology of Spaces, Lemma spaces-more-cohomology-lemma-cohomological-descent-etale-fppf. Rh_{n,*}h_n^{-1} = id on D^+(U_{n,etale}); also collapses the last step.
- Simplicial direct image. Same unit, Lemma lemma-direct-image-morphism-simplicial-sites. Rh_*h^{-1} = id on D^+(U_etale) follows levelwise.
- Descent for fppf hypercoverings. Same unit, Lemma lemma-hypercovering-X-simple-descent-bounded-abelian. Third equality: Ra_{fppf,*}a_{fppf}^{-1} = id, U an fppf hypercovering of X.

Source reference. Original reference 0DAF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5910–5910. Prerequisite pointer invoked at proof line(s) 7397; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DH5: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7233–7233. Prerequisite pointer invoked at proof line(s) 7375; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0D97: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1353–1353. Prerequisite pointer invoked at proof line(s) 7382; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D8N: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5086–5086. Prerequisite pointer invoked at proof line(s) 7398; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 7371; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DGG: sources/stacks-chow/tags.txt, source lines 16871–16871. Prerequisite pointer invoked at proof line(s) 7379; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let U be a simplicial algebraic space over S. Let $a : U \to X$ be an augmentation. If $a : U \to X$ is an fppf hypercovering of X, then for $K \in D^+(X_{étale})$*

$$K \to Ra_*(a^{-1}K)$$

is an isomorphism. Here $a : Sh(U_{étale}) \to Sh(X_{étale})$ is as in Section 0DE7.

Proof. Consider the diagram of Lemma 0DH5. Observe that $Rh_{n,*}h_n^{-1}$ is the identity functor on $D^+(U_{n,étale})$ by More on Cohomology of Spaces, Lemma 0DGG. Hence Rh_*h^{-1} is the identity functor on $D^+(U_{étale})$ by Lemma 0D97. We have

$$\begin{aligned} Ra_*(a^{-1}K) &= Ra_*Rh_*h^{-1}a^{-1}K \\ &= Rh_{-1,*}Ra_{fppf,*}a_{fppf}^{-1}(h_{-1})^{-1}K \\ &= Rh_{-1,*}(h_{-1})^{-1}K \\ &= K \end{aligned}$$

The first equality by the discussion above, the second equality because of the commutativity of the diagram in Lemma 0DAF, the third equality by Lemma 0D8N as U is a hypercovering of X in $(Spaces/S)_{fppf}$, and the last equality by the already used More on Cohomology of Spaces, Lemma 0DGG. □